Macroeconomics I — MPE Insper

Practice list: General Equilibrium, Money and Inflation (Listas 5–6)

Solutions and grading rubric

Instructions. Multiple choice: mark the single correct option (no penalty for wrong answers). *True/False*: every 2 wrong items cancel 1 right one (score = $\max\{0, \text{right} - \text{wrong}/2\}\times$ item value); blank items do not count. Discursive questions: show every step. Numbers are illustrative and chosen to give clean answers.

Block 1 — General equilibrium

(Kurlat ch. 9 • 25 points)

Question 1. (10 points) Objective items. The Tesouro asks what pins down the real rate on NTN-Bs in model economies with a representative household and competitive markets.

a) (3 points) Two-period endowment economy, $u = \ln c$, $\beta = 0.96$, $y_1 = 100$, $y_2 = 104$, no government. The equilibrium real interest rate is:

A. 4.00%

B. 8.33%

C. 4.17%

D. 0.00%

Answer: B

Step — market clearing. The representative household cannot trade with itself: $a = 0$, $c_t = y_t$. Euler: $1 + r = \dfrac{u'(y_1)}{\beta u'(y_2)} = \dfrac{y_2}{\beta y_1} = \dfrac{104}{96} = 1.0833$. Option C is $1/\beta - 1$, which ignores growth.

Ref: Kurlat (2020), §9.1, p. 165.

b) (3 points) Infinite-horizon economy on a balanced growth path where consumption grows at $g = 2\%$, $u = c^{1-\sigma}/(1-\sigma)$, $\sigma = 2$, $\beta = 0.98$. The real interest rate is:

A. 2.04%

B. 4.08%

C. 8.24%

D. 6.16%

Answer: D

Step — Euler. $(1+g)^\sigma = \beta(1+r) \Rightarrow 1 + r = 1.02^2/0.98 = 1.0616$. Approximation: $r \approx \rho + \sigma g = 2.04 + 4 = 6.04\%$.

Ref: Kurlat (2020), §9.3, p. 172.

c) (2 points, T or F) In the two-period endowment economy of (a), a rise in patience ($\beta \uparrow$) raises equilibrium saving.

Answer: F

Step — price adjusts. Aggregate saving is zero in any equilibrium. Higher β raises the desire to save at the old r, so r falls until households again want $a = 0$.

Ref: Kurlat (2020), §9.1.

d) (2 points, T or F) By the First Welfare Theorem, the steady-state capital stock of the competitive infinite-horizon economy maximizes steady-state consumption (Golden Rule).

Answer: F

Step — modified rule. The competitive steady state satisfies $F_K - \delta = 1/\beta - 1$, below the Golden Rule's $F_K - \delta = 0$. It is Pareto-efficient because impatient households value consumption now; maximizing *steady-state* consumption is not their objective.

Ref: Kurlat (2020), §9.2–9.3.

Question 2. (15 points) Discursive. Infinite-horizon economy: household with $\sum_t \beta^t c_t^{1-\sigma}/(1-\sigma)$, inelastic labor $L_t = 1$; firm with $Y = AK^\alpha L^{1-\alpha}$; capital depreciates at δ; K_0 given. Numbers: $\beta = 0.95$, $\sigma = 2$, $\alpha = 1/3$, $\delta = 0.1$, $A = 1$.

a) (5 points) Write the household's period budget constraint when it owns and rents the capital, and derive its Euler equation. Solve the firm's problem. Show that $1 + r_{t+1} = r^K_{t+1} + 1 - \delta$ and define a competitive equilibrium (allocation, prices, conditions).

Step — household. $c_t + K_{t+1} = w_t + r^K_t K_t + (1-\delta)K_t$ (plus bonds b_t in zero net supply: $+(1+r_t)b_t - b_{t+1}$).

Step — Euler. $c_t^{-\sigma} = \beta(1+r_{t+1})c_{t+1}^{-\sigma}$. The bond and capital FOCs give the same Euler only if $1 + r_{t+1} = r^K_{t+1} + 1 - \delta$ (no arbitrage).

Step — firm. $\max AK^\alpha L^{1-\alpha} - wL - r^K K$: $r^K = \alpha AK^{\alpha-1}$, $w = (1-\alpha)AK^\alpha$; CRS gives zero profits.

Step — definition. Allocation $\{c_t, K_{t+1}, L_t\}$ and prices $\{w_t, r^K_t, r_t\}$ such that (i) the household maximizes utility given prices; (ii) the firm maximizes profits given prices; (iii) markets clear: $L_t = 1$, capital demanded $= K_t$, bonds $b_t = 0$, goods $c_t + K_{t+1} - (1-\delta)K_t = AK_t^\alpha$.

Economic intuition. Bonds and capital are two ways to carry resources forward; in equilibrium they must pay the same, or no one would hold one of them.

Ref: Kurlat (2020), §9.3, pp. 172–180; Lista 5, Q2.

Grading rubric.

- Budget constraint and Euler: 1.5 pts.
- Firm FOCs: 1 pt.
- No-arbitrage derived, not just stated: 1 pt.
- Definition with all three conditions and every market: 1.5 pts. Missing market clearing: max. 3 pts in the item.

b) (5 points) Compute the steady state: r, r^K, K, Y, C, w and the saving rate $\delta K/Y$. Why does σ not appear, and why does r depend only on β?

Step — interest. $c_{t+1} = c_t \Rightarrow 1 + r = 1/\beta$: $r = 5.26\%$; $r^K = r + \delta = 0.1526$.

Step — capital. $\alpha K^{\alpha-1} = 0.1526 \Rightarrow K = (0.3333/0.1526)^{3/2} = 3.227$.

Step — rest. $Y = K^{1/3} = 1.478$; $C = Y - \delta K = 1.155$; $w = \frac{2}{3}Y = 0.985$; $\delta K/Y = 21.8\%$.

Step — σ. σ governs the willingness to deviate from flat consumption. In steady state consumption is flat, so only β matters; σ shapes the transition, not the destination.

Economic intuition. A household that keeps consumption constant needs a return that exactly compensates its impatience; technology then adjusts K until the marginal product delivers it.

Ref: Kurlat (2020), §9.3.

Grading rubric.

- r and r^K: 1.5 pts.
- K, Y, C, w, saving rate: 2.5 pts. Forgetting δ in r^K: max. 2 pts in the item.
- Role of σ and β: 1 pt.

c) (5 points) Compute the Golden Rule K_{gr} and C_{gr} and show $K < K_{gr}$. Explain why this is consistent with the First Welfare Theorem and what happens as $\beta \to 1$. Then let A rise permanently to 1.1: by how much do K, Y, C, w and r change in the new steady state?

Step — Golden Rule. $\max_K K^\alpha - \delta K \Rightarrow \alpha K^{\alpha-1} = \delta$: $K_{gr} = (0.3333/0.1)^{3/2} = 6.086$, $C_{gr} = 1.217 > 1.155$.

Step — why lower. $F_K = r + \delta > \delta$ and F_K is decreasing, so $K < K_{gr}$. Getting to K_{gr} would cost consumption now; impatient households do not want that trade. As $\beta \to 1$, $r \to 0$ and $K \to K_{gr}$.

Step — TFP. r is still $1/\beta - 1$, so r^K is unchanged and K/Y too. $K = (\alpha A/r^K)^{1/(1-\alpha)}$ scales with $A^{3/2}$: K, Y, C and w all rise by $1.1^{1.5} - 1 = 15.4\%$; r unchanged.

Economic intuition. The planner and the market agree: both discount the future at β. The Golden Rule ignores the time cost of building capital.

Ref: Kurlat (2020), §9.2 and §9.3; §4.3 (Golden Rule).

Grading rubric.

- K_{gr}, C_{gr} and the comparison: 1.5 pts.
- Consistency with the FWT (impatience) and $\beta \to 1$: 2 pts. Calling the economy inefficient: max. 2 pts in the item.
- TFP comparative statics with r unchanged: 1.5 pts.

Block 2 — Money and inflation *(Kurlat chs. 10–11 • 25 points)*

Question 3. (10 points) Objective items. Banco Central data and money-demand theory.

a) (3 points) The public holds currency equal to 25% of demand deposits and banks hold reserves equal to 15% of deposits. With a monetary base of R$ 200 billion, M1 is:

A. R$ 500 billion

B. R$ 1,333 billion

C. R$ 625 billion

D. R$ 250 billion

Answer: C

Step — multiplier. $m = \dfrac{1+cr}{rr+cr} = \dfrac{1.25}{0.40} = 3.125$; $M1 = 3.125 \cdot 200 = 625$. Option B (200/0.15) ignores currency leakage.

Ref: Kurlat (2020), §10.2–10.3, pp. 192–199.

b) (3 points) Real money demand is $m = e^{-a\pi}$ with $a = 4$ (Cagan). The steady-state inflation that maximizes revenue from the inflation tax πm is:

A. 25%

B. 40%

C. 4%

D. 100%

Answer: A

Step — Laffer. $\frac{d}{d\pi}\pi e^{-a\pi} = e^{-a\pi}(1-a\pi) = 0 \Rightarrow \pi = 1/a = 25\%$.

Ref: Kurlat (2020), §11.3, p. 215; Exercise 11.6.

c) (2 points, T or F) If money is superneutral, permanently raising money growth from 4% to 8% leaves the nominal interest rate unchanged.

Answer: F

Step — Fisher. Superneutrality concerns *real* variables: r is unchanged, so inflation rises 4 points and $i = r + \pi$ rises 4 points too (Fisher effect).

Ref: Kurlat (2020), §11.2, p. 208.

d) (2 points, T or F) With Baumol–Tobin demand $M/P = \sqrt{FY/(2i)}$ (F a real cost per trip), if real income and the nominal interest rate both double, real balances are unchanged.

Answer: T

Step — elasticities. Income elasticity $+\frac{1}{2}$, interest elasticity $-\frac{1}{2}$: the two doublings cancel inside the square root.

Ref: Kurlat (2020), §10.4, p. 199.

Question 4. (15 points) Discursive. Brazil in the 1980s. Real money demand is $M/P = \kappa Y e^{-ai}$ with $i = r + \pi$; Y grows at g, money at μ. Numbers: $\kappa = 0.3$, $a = 2.5$, $r = 0.04$, $g = 0.02$ (illustrative).

a) (5 points) Show that in a steady state with constant i, $\pi = \mu - g$. Write seigniorage $\dot{M}/P$ and the inflation tax $\pi M/P$ as shares of Y as functions of μ, and explain their difference. When does the approximation $i = r + \pi$ fail?

Step — inflation. Constant i makes $M/(PY)$ constant, so $\mu - \pi - g = 0$: $\pi = \mu - g$.

Step — seigniorage. $\dfrac{\dot{M}}{PY} = \mu \dfrac{M}{PY} = \mu\kappa e^{-a(r+\mu-g)}$; inflation tax $= \pi \dfrac{M}{PY} = (\mu - g)\kappa e^{-a(r+\mu-g)}$.

Step — difference. $g\,M/(PY)$: a growing economy demands more real balances, which the government can issue without inflation. This part is not a tax on holders.

Step — Fisher. Exact: $1 + i = (1 + r)(1 + \pi)$; the cross term $r\pi$ is negligible only at low inflation. At 80% a year it is not.

Ref: Kurlat (2020), §11.2–11.3, pp. 208–217.

Grading rubric.

- Derives $\pi = \mu - g$ from constant real balances per unit of output: 1.5 pts.
- Both revenue formulas: 1.5 pts. Treating seigniorage and inflation tax as two additive revenues: max. 2 pts in the item.
- Difference and Fisher caveat: 2 pts.

b) (5 points) Find the money growth rate that maximizes seigniorage, the associated inflation, the maximum revenue (% of GDP) and the share collected as inflation tax.

Step — FOC. $\dfrac{d}{d\mu}\mu e^{-a\mu} = e^{-a\mu}(1 - a\mu) = 0 \Rightarrow \mu^* = 1/a = 40\%$.

Step — numbers. $\pi = 38\%$; $S/Y = 0.4 \cdot 0.3 \cdot e^{-2.5(0.04+0.38)} = 0.12\, e^{-1.05} = 4.20\%$ of GDP. Inflation-tax share $= \pi/\mu = 95\%$.

Economic intuition. Above μ^* the base M/P shrinks faster than the rate rises: the Laffer curve of inflation. Money printing has a ceiling on what it can finance.

Ref: Kurlat (2020), §11.3; Exercise 11.6 (Cagan).

Grading rubric.

- FOC and $\mu^* = 1/a$: 2 pts.
- Inflation, revenue and share: 2 pts.
- Laffer interpretation: 1 pt.

c) (5 points) The government must finance a deficit of 3% of GDP by printing money. Find the steady states numerically and say which one is on the wrong side of the Laffer curve. What inflation does the Friedman rule prescribe here, and why do central banks such as the BCB target positive inflation instead?

Step — two roots. Solve $\mu \cdot 0.3\, e^{-2.5(0.02+\mu)} = 0.03$: $\mu = 15.5\%$ ($\pi = 13.5\%$) and $\mu = 82.3\%$ ($\pi = 80.3\%$). Both finance the deficit; the second is beyond $\mu^* = 40\%$.

Step — wrong side. At $\pi = 80\%$ the same revenue is raised with a much smaller base and much larger distortions (shoe-leather, menu, relative-price dispersion): strictly worse.

Step — Friedman rule. Set $i = 0$: $\pi = -r = -4\%$, removing the opportunity cost of money.

Step — why positive targets. Measurement bias in price indices, downward nominal rigidity, and room to cut i before hitting zero.

Economic intuition. A deficit that money must finance pins the inflation rate; the only choice left is which side of the Laffer curve, and expectations can push the economy to the bad one.

Ref: Kurlat (2020), §11.3–11.4, pp. 215–222.

Grading rubric.

- Both roots: 2 pts. Only one root found: max. 3 pts in the item.
- Identifies the bad equilibrium and why: 1 pt.
- Friedman rule: 1 pt.
- Reasons for positive targets: 1 pt.